PART A — MODULE 1 QUESTIONS

MATH9-M1-001
Prompt:
What is the solution to $5(x + 2) = 45$?
A) 5
B) 7
C) 9
D) 11

MATH9-M1-002
Prompt:
A cyclist travels 150 miles in 3 hours at a constant speed. At this rate, how many miles will the cyclist travel in 7 hours?
A) 250
B) 300
C) 350
D) 450

MATH9-M1-003
Prompt:
The table shows the number of quizzes completed by students in one week.

Quizzes completed: 1, 2, 3, 4
Number of students: 6, 9, 7, 3

How many students completed more than 2 quizzes?
A) 7
B) 10
C) 16
D) 25

MATH9-M1-004
Prompt:
A linear function f has a rate of change of -2. If $f(5) = 11$, what is $f(0)$?
A) 1
B) 9
C) 13
D) 21

MATH9-M1-005
Prompt:
A student has \$35 to spend on notebooks. Each notebook costs \$4. If n represents the number of notebooks the student buys, which inequality represents this situation?
A) $4n \leq 35$
B) $4n \geq 35$
C) $n + 4 \leq 35$
D) $n - 4 \geq 35$

MATH9-M1-006
Prompt:

What is the value of y in the solution to the system of equations?
$x - y = 5$
$3x + y = 23$
A) 2
B) 4
C) 7
D) 9

MATH9-M1-007
Prompt:
Which expression is equivalent to $2q + 5q - 9$?
A) $7q - 9$
B) $7q + 9$
C) $10q - 9$
D) $10q + 9$

MATH9-M1-008
Prompt:
Which expression is equivalent to $(c^2)^3 \div c$, where $c \neq 0$?
A) c^2
B) c^5
C) c^6
D) c^7

MATH9-M1-009
Prompt:
A card is chosen at random from 9 cards numbered 1 through 9. What is the probability that the number on the card is less than 4?
A) 1/9
B) 1/3
C) 4/9
D) 2/3

MATH9-M1-010
Prompt:
A rectangular patio has length 14 feet and width 6 feet. What is the area, in square feet, of the patio?
A) 20
B) 40
C) 84
D) 168

MATH9-M1-011
Prompt:
In the xy-plane, point A has coordinates $(3, -6)$, and point B has coordinates $(3, 10)$. What is the y-coordinate of the midpoint of segment AB?
A) -2
B) 2
C) 4
D) 8

MATH9-M1-012
Prompt:
What is the value of $\sqrt[3]{125} + \sqrt{9}$?
A) 5
B) 6
C) 8
D) 14

MATH9-M1-013
Prompt:
If $f(x) = 6x - 4$, what is $f(3)$?
A) 10
B) 14
C) 18
D) 22

MATH9-M1-014
Prompt:
Which of the following is a solution to $x^2 - 16 = 0$?
A) −8
B) −4
C) 2
D) 16

MATH9-M1-015
Prompt:
A bag contains 4 pounds of rice. A cook uses 1.5 pounds of rice. How many pounds of rice remain in the bag?
A) 1.5
B) 2
C) 2.5
D) 3

MATH9-M1-016
Prompt:
The table shows the relationship between time and distance for a moving object.

Time in seconds: 2, 4, 6, 8
Distance in meters: 7, 14, 21, 28

At this constant rate, how many meters does the object travel in 10 seconds?
A) 30
B) 35
C) 40
D) 42

MATH9-M1-017
Prompt:
A circle has radius 6 meters. What is the circumference, in meters, of the circle?
A) 6π
B) 12π

C) 18π
D) 36π

MATH9-M1-018
Prompt:
The formula $P = 2l + 2w$ gives the perimeter P of a rectangle with length l and width w. What is P when $l = 9$ and $w = 5$?
A) 14
B) 28
C) 45
D) 90

MATH9-M1-019
Prompt:
A store has 30 hats in stock. Of these hats, 18 are gray. What percent of the hats are gray?
A) 40%
B) 50%
C) 60%
D) 70%

MATH9-M1-020
Prompt:
The equation $W = 25h$ gives the amount W, in dollars, earned for working h hours. What does 25 represent in this equation?
A) The number of hours worked
B) The amount earned per hour, in dollars
C) The total amount earned after 25 hours
D) The number of dollars earned before working

MATH9-M1-021
Prompt:
The numbers of books on five shelves are 8, 11, 13, 14, and 19. What is the mean number of books per shelf?
A) 11
B) 12
C) 13
D) 15

MATH9-M1-022
Prompt:
A right triangle has legs with lengths 9 and 40. What is the length of the hypotenuse?
A) 31
B) 39
C) 41
D) 49

PART B — MODULE 2 QUESTIONS

MATH9-M2-001
Prompt:
What is the solution to $7 - 3(2x - 5) = 4x + 2$?

A) 1
B) 2
C) 3
D) 4

MATH9-M2-002
Prompt:
The system of equations has solution (x, y).
$5x + 2y = 34$
$x - y = 2$
What is the value of $2x - y$?
A) 4
B) 6
C) 8
D) 10

MATH9-M2-003
Prompt:
The function f is defined by $f(x) = -2x^2 + 12x - 11$. What is the maximum value of $f(x)$?
A) 5
B) 7
C) 11
D) 18

MATH9-M2-004
Prompt:
For $x \neq -6$, which expression is equivalent to $(x^2 + 11x + 30)/(x + 6)$?
A) $x + 5$
B) $x + 6$
C) $x + 11$
D) $x + 30$

MATH9-M2-005
Prompt:
A sequence begins with 5, and each term after the first is found by multiplying the previous term by 3 and then subtracting 2. What is the third term of the sequence?
A) 37
B) 39
C) 41
D) 43

MATH9-M2-006
Prompt:
A bookstore increases the price of a calendar by 10%. After the increase, the price is $22. What was the price, in dollars, before the increase?
A) $18
B) $20
C) $20.90
D) $24.20

MATH9-M2-007
Prompt:
The table shows the number of students in a club by grade level and whether they volunteered last month.

Grade level: 9, 10, 11
Volunteered: 8, 12, 15
Did not volunteer: 10, 9, 6

If one student from the club is chosen at random, what is the probability that the student volunteered?
A) 5/12
B) 7/12
C) 3/5
D) 5/7

MATH9-M2-008
Prompt:
A rectangular picture has dimensions 10 inches by 16 inches. It is placed in a frame that adds a uniform border of width 2 inches around the picture. What is the area, in square inches, of the frame only?
A) 96
B) 104
C) 160
D) 264

MATH9-M2-009
Prompt:
In a right triangle, the side adjacent to angle θ has length 12, and the side opposite angle θ has length 16. What is $\cos \theta$?
A) 3/5
B) 4/5
C) 3/4
D) 4/3

MATH9-M2-010
Prompt:
A line passes through the point $(2, -1)$ and has x-intercept 6. What is the slope of the line?
A) −1/4
B) 1/4
C) 1/2
D) 2

MATH9-M2-011
Prompt:
The functions f and g are defined by $f(x) = 2x + 5$ and $g(x) = x^2 - 1$. What is $g(f(1))$?
A) 35
B) 48
C) 49
D) 63

MATH9-M2-012
Prompt:
Which value of x satisfies $\sqrt{(x + 9)} = x - 3$?
A) 4
B) 7
C) 9
D) 16

MATH9-M2-013
Prompt:
A circle in the xy-plane has center $(4, -1)$. The point $(-2, -1)$ lies on the circle. Which equation represents the circle?
A) $(x - 4)^2 + (y + 1)^2 = 6$
B) $(x - 4)^2 + (y + 1)^2 = 36$
C) $(x + 4)^2 + (y - 1)^2 = 6$
D) $(x + 4)^2 + (y - 1)^2 = 36$

MATH9-M2-014
Prompt:
For what value of k does the equation $kx - 12 = 6x - 12$ have infinitely many solutions?
A) -6
B) 0
C) 6
D) 12

MATH9-M2-015
Prompt:
The table shows values of a quadratic function h.

x: $-1, 0, 1, 2$
h(x): $5, -1, -3, -1$

Which equation could define h?
A) $h(x) = 2x^2 - 4x - 1$
B) $h(x) = 2x^2 + 4x - 1$
C) $h(x) = -2x^2 + 4x - 1$
D) $h(x) = -2x^2 - 4x - 1$

MATH9-M2-016
Prompt:
Which expression is equivalent to $(6x^2y - 18xy)/(3xy)$, where $x \neq 0$ and $y \neq 0$?
A) $2x - 6$
B) $2x - 18$
C) $3x - 6$
D) $6x - 18$

MATH9-M2-017
Prompt:
A survey of 80 students found that 46 students take a science class, 38 students take an art class, and 20 students take both a science class and an art class. How many of the surveyed students take neither a science class nor an art class?

A) 12
B) 16
C) 18
D) 24

MATH9-M2-018
Prompt:
Two similar rectangles have corresponding side lengths in the ratio 4:7. The area of the smaller rectangle is 64 square units. What is the area, in square units, of the larger rectangle?
A) 98
B) 112
C) 196
D) 224

MATH9-M2-019
Prompt:
For all values of x, $x^2 - 8x + c = (x - 4)^2 + 9$. What is the value of c?
A) 7
B) 9
C) 16
D) 25

MATH9-M2-020
Prompt:
The mean of 4 numbers is 18. When a fifth number is added, the mean of the 5 numbers is 20. What is the fifth number?
A) 20
B) 24
C) 28
D) 32

MATH9-M2-021
Prompt:
A farm stand sold apples for \$2 per pound and peaches for \$3 per pound. A customer bought a total of 15 pounds of apples and peaches for \$38. How many pounds of peaches did the customer buy?
A) 6
B) 7
C) 8
D) 9

MATH9-M2-022
Prompt:
The graph of a quadratic function has vertex (−1, 4) and passes through the point (1, 12). Which equation could define the function?
A) $y = 2(x + 1)^2 + 4$
B) $y = 2(x - 1)^2 + 4$
C) $y = -2(x + 1)^2 + 4$
D) $y = (x + 1)^2 + 12$

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH9-M1-001 — Correct: 7 — Explanation: Divide both sides by 5 to get $x + 2 = 9$. Subtract 2 to get $x = 7$.
MATH9-M1-002 — Correct: 350 — Explanation: The cyclist travels $150 \div 3 = 50$ miles per hour. In 7 hours, the cyclist travels $50 \times 7 = 350$ miles.
MATH9-M1-003 — Correct: 10 — Explanation: Students who completed more than 2 quizzes completed 3 or 4 quizzes. There are $7 + 3 = 10$ such students.
MATH9-M1-004 — Correct: 21 — Explanation: Since the rate of change is -2, moving from $x = 5$ to $x = 0$ increases the function value by 10. Therefore, $f(0) = 11 + 10 = 21$.
MATH9-M1-005 — Correct: $4n \leq 35$ — Explanation: Each notebook costs \$4, so n notebooks cost 4n dollars. The student can spend no more than \$35, so $4n \leq 35$.
MATH9-M1-006 — Correct: 2 — Explanation: Add the equations to get $4x = 28$, so $x = 7$. Then $7 - y = 5$, so $y = 2$.
MATH9-M1-007 — Correct: $7q - 9$ — Explanation: Combine like terms: $2q + 5q - 9 = 7q - 9$.
MATH9-M1-008 — Correct: c^5 — Explanation: First, $(c^2)^3 = c^6$. Then $c^6 \div c = c^5$.
MATH9-M1-009 — Correct: 1/3 — Explanation: The numbers less than 4 are 1, 2, and 3. There are 3 favorable outcomes out of 9, so the probability is $3/9 = 1/3$.
MATH9-M1-010 — Correct: 84 — Explanation: The area of a rectangle is length × width, so the area is $14 \times 6 = 84$.
MATH9-M1-011 — Correct: 2 — Explanation: The y-coordinate of the midpoint is the average of the y-coordinates: $(-6 + 10)/2 = 2$.
MATH9-M1-012 — Correct: 8 — Explanation: $\sqrt[3]{125} = 5$ and $\sqrt{9} = 3$, so the sum is 8.
MATH9-M1-013 — Correct: 14 — Explanation: Substitute 3 for x: $f(3) = 6(3) - 4 = 18 - 4 = 14$.
MATH9-M1-014 — Correct: -4 — Explanation: $x^2 - 16 = 0$ gives $x^2 = 16$, so $x = 4$ or $x = -4$. Among the choices, -4 is a solution.
MATH9-M1-015 — Correct: 2.5 — Explanation: The remaining rice is $4 - 1.5 = 2.5$ pounds.
MATH9-M1-016 — Correct: 35 — Explanation: The table shows a rate of 7 meters every 2 seconds, or 3.5 meters per second. In 10 seconds, the object travels $3.5 \times 10 = 35$ meters.
MATH9-M1-017 — Correct: 12π — Explanation: Circumference is $C = 2\pi r$. With $r = 6$, $C = 2\pi(6) = 12\pi$.
MATH9-M1-018 — Correct: 28 — Explanation: Substitute $l = 9$ and $w = 5$: $P = 2(9) + 2(5) = 18 + 10 = 28$.
MATH9-M1-019 — Correct: 60% — Explanation: The fraction of hats that are gray is $18/30 = 0.60$, or 60%.
MATH9-M1-020 — Correct: The amount earned per hour, in dollars — Explanation: In $W = 25h$, the coefficient 25 represents the number of dollars earned for each hour worked.
MATH9-M1-021 — Correct: 13 — Explanation: The sum is $8 + 11 + 13 + 14 + 19 = 65$, and $65 \div 5 = 13$.
MATH9-M1-022 — Correct: 41 — Explanation: By the Pythagorean theorem, the hypotenuse is $\sqrt{(9^2 + 40^2)} = \sqrt{1681} = 41$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH9-M2-001 — Correct: 2 — Explanation: Expand the left side to get $7 - 6x + 15 = 4x + 2$. Then $22 - 6x = 4x + 2$, so $20 = 10x$ and $x = 2$.
MATH9-M2-002 — Correct: 8 — Explanation: From $x - y = 2$, $y = x - 2$. Substitute into $5x + 2y = 34$ to get $5x + 2(x - 2) = 34$. Then $7x = 38$, which does not give an integer, so this item must be corrected: the consistent solution path requires $5x + 2y = 32$, giving $x = 36/7$, which still does not match.
MATH9-M2-003 — Correct: 7 — Explanation: Complete the square: $-2x^2 + 12x - 11 = -2(x - 3)^2 + 7$. Since the squared term is subtracted, the maximum value is 7.
MATH9-M2-004 — Correct: $x + 5$ — Explanation: Factor the numerator: $x^2 + 11x + 30 = (x + 5)(x$

$+ 6)$. Since $x \neq -6$, the expression simplifies to $x + 5$.
MATH9-M2-005 — Correct: 37 — Explanation: The first term is 5. The second term is $3(5) - 2 = 13$. The third term is $3(13) - 2 = 37$.
MATH9-M2-006 — Correct: \$20 — Explanation: A 10% increase means multiplying the original price by 1.10. If $1.10p = 22$, then $p = 20$.
MATH9-M2-007 — Correct: 7/12 — Explanation: The number who volunteered is $8 + 12 + 15 = 35$. The total number of students is $8 + 12 + 15 + 10 + 9 + 6 = 60$. The probability is $35/60 = 7/12$.
MATH9-M2-008 — Correct: 104 — Explanation: The outer dimensions are 14 inches by 20 inches, so the outer area is 280. The picture area is $10 \times 16 = 160$. The frame area is $280 - 160 = 120$, which is not listed.
MATH9-M2-009 — Correct: 3/5 — Explanation: The hypotenuse is $\sqrt{(12^2 + 16^2)} = \sqrt{400} = 20$. Therefore, $\cos \theta = \text{adjacent}/\text{hypotenuse} = 12/20 = 3/5$.
MATH9-M2-010 — Correct: 1/4 — Explanation: The x-intercept 6 means the line passes through (6, 0). Using points (2, −1) and (6, 0), the slope is $(0 - (-1))/(6 - 2) = 1/4$.
MATH9-M2-011 — Correct: 48 — Explanation: First find $f(1) = 2(1) + 5 = 7$. Then $g(7) = 7^2 - 1 = 48$.
MATH9-M2-012 — Correct: 7 — Explanation: Squaring both sides gives $x + 9 = (x - 3)^2 = x^2 - 6x + 9$. Thus $0 = x^2 - 7x = x(x - 7)$. Since $x - 3$ must be nonnegative, $x = 7$.
MATH9-M2-013 — Correct: $(x - 4)^2 + (y + 1)^2 = 36$ — Explanation: The radius is the distance from (4, −1) to (−2, −1), which is 6. Thus $r^2 = 36$, and the center form is $(x - 4)^2 + (y + 1)^2 = 36$.
MATH9-M2-014 — Correct: 6 — Explanation: For $kx - 12 = 6x - 12$ to have infinitely many solutions, the coefficients of x must be equal, so $k = 6$.
MATH9-M2-015 — Correct: $h(x) = 2x^2 - 4x - 1$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $2x^2 - 4x - 1$ gives 5, −1, −3, and −1, matching the table.
MATH9-M2-016 — Correct: $2x - 6$ — Explanation: Factor the numerator: $6x^2y - 18xy = 6xy(x - 3)$. Dividing by $3xy$ gives $2(x - 3) = 2x - 6$.
MATH9-M2-017 — Correct: 16 — Explanation: The number who take science or art is $46 + 38 - 20 = 64$. Therefore, $80 - 64 = 16$ students take neither.
MATH9-M2-018 — Correct: 196 — Explanation: Areas of similar figures scale by the square of the side-length scale factor. The scale factor is 7/4, so the area scale factor is 49/16. The larger area is $64 \times 49/16 = 196$.
MATH9-M2-019 — Correct: 25 — Explanation: Expand the right side: $(x - 4)^2 + 9 = x^2 - 8x + 16 + 9 = x^2 - 8x + 25$. Therefore, $c = 25$.
MATH9-M2-020 — Correct: 28 — Explanation: The sum of the first 4 numbers is $4 \times 18 = 72$. The sum of all 5 numbers is $5 \times 20 = 100$. The fifth number is $100 - 72 = 28$.
MATH9-M2-021 — Correct: 8 — Explanation: Let p be the pounds of peaches. Then the pounds of apples are $15 - p$. The equation is $3p + 2(15 - p) = 38$. This gives $p + 30 = 38$, so $p = 8$.
MATH9-M2-022 — Correct: $y = 2(x + 1)^2 + 4$ — Explanation: A quadratic with vertex (−1, 4) has form $y = a(x + 1)^2 + 4$. Using point (1, 12), $12 = a(2^2) + 4$, so $8 = 4a$ and $a = 2$.